



Grade 3 Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADE 3 · FULL YEAR

160 min per week · 2 lessons weekly

About this overview. This document is an example of how the course might be taught. More investigations and examples are approved for this course than appear here, so scholars may see some variation in activities and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Grade 3 Science explores the natural world through the lens of America's national parks and Florida's own ecosystems. The year runs as 33 weekly modules, each with two lessons: the first introduces a core idea, and the second is an investigation, lab, data task, or engineering challenge. Scholars keep an observation notebook, graph real data, build and test designs, and finish the year researching a national park ecosystem of their own.

THE YEAR AT A GLANCE

1 Science and Observing (Module 1)

Becoming a park scientist: observation versus inference, scientist tools, and keeping a science notebook.

3 Forces and Motion (Modules 5–7)

Pushes and pulls, predictable motion, and magnets, with ramp, pendulum, and magnetic-rescue investigations.

5 Adaptations and Survival (Modules 12–16)

Habitats, ecosystems, food chains, limiting factors, and adaptations, capped by designing an adapted animal.

7 Plants and Their Environments (Modules 18–21)

Plant survival strategies, photosynthesis, plant responses, and reproduction.

9 Engineering and Florida Ecosystems (Modules 26–31)

The engineering design process applied to water and wildlife, then the Everglades, Florida wildlife, and Florida's environmental challenges.

2 Weather, Climate, and Earth Systems (Modules 2–4)

Weather patterns, climates across national parks, natural hazards, and a storm-safe structure design challenge.

4 Life Cycles, Traits, and Inheritance (Modules 8–11)

Plant and animal life cycles, inherited traits, and how variation helps organisms survive.

6 Fossils and Earth History (Module 17)

What fossils tell us about past life, with a fossil-dig simulation.

8 Human Impact and Conservation (Modules 22–25)

Pollution, conservation, and why we protect national parks; scholars draft a park-protection proposal.

10 Synthesis: National Park Research (Modules 32–33)

A national park research project and a final investigation connecting the year's big ideas.

A TYPICAL WEEK

- Two science lessons each week, completing one module.
- The first lesson introduces the week's core idea with readings, examples, and guided observation.
- The second is hands-on: an investigation, lab, data task, or engineering challenge where scholars collect evidence and explain results.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

TIME COMMITMENT

160 minutes / week

two lessons a week, about 80 minutes each

- Each module spans one week: one concept lesson, one investigation.
- Engineering challenges and the final research project may run a little longer.

WHAT SCHOLARS NEED

- A science notebook for observations, data tables, and labeled drawings.
- Simple household materials for investigations and design challenges.
- A computer with reliable internet and access to Canvas.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting twice a week on the school's weekly schedule.